

Machine Learning Lab (BCSL606) –Viva Questions with Answers

PROGRAM 1 – Histograms and Boxplots (California Housing Dataset)

1. What is a histogram?

A histogram is a graphical representation used to show the frequency distribution of numerical data. It helps us understand how the values are spread in the dataset.

2. Why do we use boxplots?

Boxplots are used to identify the spread of data and detect outliers. They display minimum value, maximum value, median, and quartiles.

3. What are outliers?

Outliers are the data points that are very far away from the normal range of values. They may affect the performance of machine learning models.

4. Why is data visualization important?

Data visualization helps us understand patterns, trends, and distributions in the dataset before applying machine learning algorithms.

5. Which libraries are used in this program?

Mainly Pandas, Matplotlib, and Seaborn libraries are used for handling and visualizing data.

PROGRAM 2 – Correlation Matrix and Pair Plot

1. What is correlation?

Correlation shows the relationship between two variables. It tells whether variables move together positively or negatively.

2. What is the range of correlation coefficient?

The correlation coefficient ranges from -1 to +1.

+1 means strong positive correlation

-1 means strong negative correlation

0 means no correlation

3. What is a heatmap?

A heatmap is a graphical representation of the correlation matrix using colors to indicate the strength of correlation.

4. What is pairplot?

Pairplot is used to visualize pairwise relationships between multiple features in a dataset.

5. Why do we analyze correlation?

Correlation analysis helps in feature selection and understanding dependencies between variables.

PROGRAM 3 – PCA (Principal Component Analysis)

1. What is PCA?

PCA is a dimensionality reduction technique used to reduce the number of features while preserving important information.

2. Why is PCA used?

PCA reduces computational complexity and helps in visualizing high-dimensional data.

3. What are principal components?

Principal components are new variables formed by combining original features in such a way that maximum variance is retained.

4. Is PCA supervised or unsupervised?

PCA is an unsupervised learning technique because it does not use target labels.

5. Why is Iris dataset commonly used?

The Iris dataset is simple, clean, and widely used for understanding classification and dimensionality reduction concepts.

PROGRAM 4 – Find-S Algorithm

1. What is Find-S algorithm?

Find-S is a concept learning algorithm used to find the most specific hypothesis that fits all positive training examples.

2. What type of learning is Find-S?

It is a supervised learning algorithm.

3. Does Find-S consider negative examples?

No. Find-S only considers positive examples.

4. What is hypothesis in machine learning?

A hypothesis is a proposed explanation or rule that maps inputs to outputs.

5. What is the limitation of Find-S?

It ignores negative examples and may not work properly with noisy data.

PROGRAM 5 – K-Nearest Neighbour (KNN)

1. What is KNN?

KNN is a supervised learning algorithm used for classification and regression based on nearest data points.

2. What does “K” represent?

K represents the number of nearest neighbors considered for prediction.

3. How does KNN classify data?

KNN calculates the distance between points and assigns the class based on majority voting.

4. Which distance measure is commonly used?

Euclidean distance is commonly used in KNN.

5. What happens if K value is too small?

The model may become sensitive to noise and overfit the data.

PROGRAM 6 – Locally Weighted Regression

1. What is Locally Weighted Regression?

It is a non-parametric regression algorithm that fits a curve locally around the query point.

2. Why is it called non-parametric?

Because it does not assume a fixed form of the model.

3. What is the role of bandwidth parameter?

Bandwidth controls how much influence nearby points have on prediction.

4. What is regression?

Regression is used to predict continuous numerical values.

5. What is the advantage of locally weighted regression?

It can model complex and nonlinear relationships effectively.

PROGRAM 7 – Linear Regression and Polynomial Regression

1. What is Linear Regression?

Linear Regression is used to establish a linear relationship between independent and dependent variables.

2. What is Polynomial Regression?

Polynomial Regression models nonlinear relationships by adding polynomial terms.

3. What is the equation of linear regression?

The equation is: $y = mx + c$

where

m = slope

c = intercept

4. What is overfitting in polynomial regression?

When the model fits training data too closely and performs poorly on new data, it is called overfitting.

5. Which dataset is used for polynomial regression in this experiment?

Auto MPG dataset is commonly used.

PROGRAM 8 – Decision Tree Algorithm

1. What is a Decision Tree?

A Decision Tree is a supervised learning algorithm used for classification and regression.

2. What are nodes in decision tree?

Nodes represent conditions or decisions based on feature values.

3. What is root node?

The topmost node of the tree is called the root node.

4. What is entropy?

Entropy measures the impurity or randomness in the dataset.

5. What is information gain?

Information gain measures how well a feature separates the data into classes.

PROGRAM 9 – Naive Bayes Classifier

1. What is Naive Bayes algorithm?

Naive Bayes is a probabilistic classifier based on Bayes theorem.

2. Why is it called “naive”?

Because it assumes all features are independent of each other.

3. What is Bayes theorem?

Bayes theorem calculates conditional probability using prior knowledge.

4. What are the advantages of Naive Bayes?

It is simple, fast, and works well for large datasets.

5. Where is Naive Bayes commonly used?

It is commonly used in spam filtering, text classification, and face recognition.

PROGRAM 10 – K-Means Clustering

1. What is K-Means clustering?

K-Means is an unsupervised learning algorithm used to group similar data points into clusters.

2. What does “K” represent?

K represents the number of clusters.

3. What is centroid?

Centroid is the center point of a cluster.

4. Is K-Means supervised or unsupervised?

K-Means is an unsupervised learning algorithm.

5. What is the main objective of K-Means?

The objective is to minimize the distance between data points and their cluster centroids.

